

Original Manuscript ID: TSC-2025-01-0005

Original Article Title: “Multi-Objective DNN Inference Task Deployment for Reliability and Delay”

To: IEEE TSC Editor

Re: Response to reviewers

Dear Editor,

We thank you and the reviewers for the constructive comments and for the opportunity to revise our manuscript. We have carefully studied all comments and have substantially revised the manuscript in terms of system modeling, algorithm design, experimental evaluation, related work, and writing quality.

In the revised manuscript, the deployment problem is reformulated for dynamic cloud-edge-end DNN inference. The revised model considers DAG-structured inference dependencies, time-varying node and link states, dynamic service reliability, inference accuracy, inference delay, and total energy consumption. We also redesign the algorithm as **DAREED: DAG-Aware Reliable and Energy-Efficient Evolutionary Deployment algorithm**, which incorporates DAG-aware evolutionary operators, resource-aware variation, constraint-aware selection, elite refinement, and preference-based deployment selection. The performance analysis section has been expanded with parameter sensitivity, Pareto quality, overall online deployment performance, **cloud-side stress tests, dynamic degradation tests**, ablation studies, scalability evaluation, and application preference scenarios.

The comments are reproduced below, followed by our point-by-point responses and corresponding actions.

Best regards,

Ying Wang, Huang Lianze, Wei Junjie et al.

Response to Associate Editor

Associate Editor Comment: The manuscript requires major revision. The key concerns include insufficient algorithmic novelty, unrealistic modeling assumptions, unclear mathematical formulations, weak experimental evaluation, and writing quality issues. The authors are also encouraged to consider additional optimization objectives such as energy consumption and cost, and to test the framework under larger and more dynamic network conditions.

Author response:

Thanks for the editor's summary and guidance. We have conducted a major revision to address these issues systematically.

First, the system model has been redesigned from a static node-reliability model to a dynamic cloud-edge-end inference model. The revised model captures online DNN request arrivals, **running task blocks**, time-varying node and link load, heat accumulation, dynamic node reliability, dynamic link reliability, inference accuracy, critical-path inference delay, and total energy consumption.

Second, the algorithm has been redesigned and renamed as DAREED. Rather than directly applying a generic NSGA-II framework, DAREED embeds DNN DAG information and dynamic resource states into initialization, crossover, mutation, repair, constraint-dominance sorting, elite local search, and final preference-based selection.

Third, the experimental design has been expanded. The revised performance analysis includes state-of-the-art and stronger baselines, repeated random-seed experiments with standard deviation reporting, Pareto-front quality metrics, scalability tests, dynamic degradation tests, cloud-side stress tests, and ablation studies.

Author action:

We revised the Abstract, Introduction, Motivation, Problem Formulation, Algorithm Design, Performance Analysis, Related Work, and Conclusion sections accordingly. We also proofread the manuscript to improve terminology consistency and writing quality.

Response to Reviewer #1

Reviewer#1, Concern#1: In the Abstract, the manuscript claims to use NSGA-II “to optimize both reliability and delay”, but it then explicitly lists service reliability, accuracy reliability, and delay as optimization objectives, which is effectively a three-objective formulation. Without clarifying that “reliability” is decomposed into service and accuracy aspects, this wording may appear internally inconsistent and weaken the clarity of the main contribution.

Author response:

Thanks for pointing out this inconsistency. We have revised the terminology throughout the manuscript. The term “accuracy reliability” has been replaced by **inference accuracy**, and the optimization objectives are now consistently stated as **service reliability, inference accuracy, inference delay, and total energy consumption**. This avoids mixing service availability and inference quality under an ambiguous reliability term.

Author action:

We revised the Abstract, Introduction, Problem Formulation, Algorithm Design, and Performance Analysis sections to use the four objective terms consistently.

Reviewer#1, Concern#2: In Related Work, the manuscript uses strongly judgmental phrasing such as “relying on fixed, hand-designed reward weights is outdated” and suggests that preference-driven systems can “select the perfect deployment strategy instantly”. This tone reads as subjective and promotional.

Author response:

Thanks for the suggestion. We agree that the previous wording was overly promotional. The revised Related Work section now uses neutral, evidence-based language. It discusses the limitations of fixed weighted aggregation in dynamic DNN inference deployment without using subjective expressions.

Author action:

We rewrote the Related Work discussion on multi-objective optimization and preference-based decision making. The revised text emphasizes that fixed weights are difficult to determine under heterogeneous request preferences and time-varying system states.

Reviewer#1, Concern#3: In Equation (6), task service reliability is defined as a product over all deployed layers and nodes. This implicitly assumes independent failures and effectively multiplies reliability for each layer execution. For models with many layers, the value can become extremely small even if each node is highly reliable.

Author response:

Thanks for identifying this modeling weakness. We have reformulated service reliability to avoid repeatedly multiplying the reliability of the same node for multiple layers deployed on that node. In the revised model, the service reliability of a DNN request is calculated over the **unique computing nodes** and **unique communication links** used by the deployment strategy. This formulation better reflects the fact that when several subtasks are placed on the same node, a node failure affects them jointly rather than independently.

Author action:

We revised the reliability model in Section 4. The revised service reliability model includes both node reliability and link reliability, and it is defined over the sets of used nodes and traversed links.

Reviewer#1, Concern#4: The manuscript contains several copy-editing and grammar issues that affect readability.

Author response:

Thanks for the careful reading. We have performed a comprehensive proofreading pass. We corrected capitalization in figure captions, grammar errors in the reliability discussion, and inaccurate expressions in the performance analysis. We also unified the use of “end device”, “service reliability”, “inference accuracy”, “inference delay”, and “total energy consumption”.

Author action:

We revised the wording and punctuation throughout the manuscript, including figure captions, formula descriptions, objective definitions, and experimental discussions.

Reviewer#1, Concern#5: The reliability model assumes that node reliability parameters remain constant during task execution. However, in the experiments, reliability values are randomly drawn from intervals. It is unclear whether these values are fixed per node across all tasks or re-sampled across runs.

Author response:

Thanks for pointing out the ambiguity. The revised model explicitly distinguishes initial service quality from dynamic service quality. Each node has an initial service reliability and initial inference accuracy sampled at the beginning of a simulation instance. During online execution, these values evolve according to the current node load and heat state.

Author action:

We revised Section 4 to define dynamic node reliability and dynamic inference accuracy. We also revised Section 6 to state that experiments are repeated under different random seeds and that mean and standard deviation are reported.

Reviewer#1, Concern#6: In Equations (11) and (12), service reliability and accuracy reliability are formulated as products over all nodes and layers using the binary variable. Since each layer is deployed on exactly one node, the current product form implicitly includes reliability terms of non-selected nodes.

Author response:

Thanks for the comment. We have reformulated both objectives. Service reliability is now computed from the unique nodes and links used by the deployment. Inference accuracy is now modeled as a workload-weighted inference quality over the nodes actually selected by the deployment strategy. Thus, non-selected nodes do not participate in the objective evaluation.

Author action:

We revised the mathematical formulation of service reliability and inference accuracy in Section 4 and updated the notation table accordingly.

Revised manuscript text:

Reviewer#1, Concern#7: The RDODA algorithm retains delay-violating solutions by assigning negative fitness values in the delay direction, instead of removing them. The manuscript does not explain how negative values affect non-dominated sorting.

Author response:

Thanks for the suggestion. The revised DAREED algorithm no longer relies on an unexplained negative-fitness treatment. Instead, it uses a constraint-aware evaluation mechanism. Delay violation, CPU violation, and path feasibility violation are accumulated into a constraint violation value. Feasible individuals are preferred over infeasible individuals; among infeasible individuals, those with smaller violations are preferred. This avoids distorting Pareto dominance while still preserving useful search information in early generations.

Author action:

We revised Section 5 to include constraint-aware repair, constraint violation calculation, and constraint-dominance sorting.

Reviewer#1, Concern#8: Constraints sum over all tasks and layers, suggesting joint deployment across multiple tasks. Meanwhile, objective functions are written with a single-task form, which can confuse whether the optimization is per-task or multi-task.

Author response:

Thanks for pointing out this ambiguity. We have clarified that the deployment process is online. DAREED optimizes the deployment of the newly arrived DNN request under the current system state. Previously accepted requests are modeled as running blocks that occupy node and link resources for several time slots. Therefore, the current request is optimized individually, while global resource constraints are affected by all currently running requests.

Author action:

We revised Section 4 to introduce the running-block model and dynamic state transition constraint. We also revised Section 5 to explain that DAREED is executed for each newly arrived request under the current resource state.

Reviewer#1, Concern#9: The simulation assumes the central cloud has unlimited resources and high service reliability. Adding stress tests with constrained cloud compute, reduced cloud reliability, or tighter cloud-edge bandwidth would better demonstrate the reliability-delay trade-off.

Author response:

Thanks for the valuable suggestion. We have added a cloud-side stress test to evaluate whether DAREED depends on idealized cloud conditions. The revised experiment includes constrained cloud compute, degraded cloud reliability and inference accuracy, and narrowed edge-cloud bandwidth. This setting is designed to show how DAREED shifts deployment decisions among cloud, edge, and end resources when the cloud is no longer uniformly advantageous.

Author action:

We added a “Cloud Constraint Stress Test” subsection in Section 6, including the planned figure and table for service reliability, inference accuracy, inference delay, total energy consumption, and failed deployments.

Reviewer#1, Concern#10: The manuscript claims the highest hypervolume value, but does not clarify whether the reported hypervolume values are directly comparable across algorithms under identical normalization and reference point settings.

Author response:

Thanks for the comment. We have clarified the hypervolume calculation procedure. In the revised evaluation, Pareto-front quality is measured in the normalized objective space of service reliability, inference accuracy, and delay satisfaction. All

algorithms use the same normalization method and the same reference point. The final hypervolume is averaged over DNN requests, and standard deviation is reported.

Author action:

We revised the Metrics and Comparison Methods subsection in Section 6 to define the normalized objective space and the hypervolume reference point.

Reviewer#1, Concern#11: Many key parameters are randomly selected within ranges, but the manuscript does not specify how many random instances were generated, whether results are averaged over multiple seeds, or what variability measure is used.

Author response:

Thanks for the suggestion. We have revised the experimental setup to state that experiments are repeated under different random seeds. The revised result tables are designed to report mean values and standard deviations.

Author action:

We revised Section 6 to include repeated experiments, random-seed settings, and standard-deviation reporting in Pareto quality and performance tables.

Reviewer#1, Concern#12: Baselines LB and LPD are single-policy heuristics rather than multi-objective methods. The manuscript could better contextualize percentage improvements and discuss limitations of these baselines.

Author response:

Thanks for the suggestion. We have expanded the comparison methods. The revised experiments include not only simple heuristic baselines, but also a generic NSGA-II based evolutionary deployment algorithm, simulated annealing, random feasible deployment, maximum-resource deployment, local-first deployment, and RTBL, a reliability-aware task scheduling method based on Lyapunov optimization and bandit learning. We also clarify that single-policy heuristics are included as interpretable references rather than as complete multi-objective competitors.

Author action:

We revised Section 6 to add a comparison algorithm table and to separate Pareto-front quality evaluation from final online deployment performance evaluation.

Response to Reviewer #2

Reviewer#2, Concern#1: The authors can include the contributions of recent papers and provide a tabular comparative analysis between their work and previous research, highlighting the unique contribution of their paper.

tabular

Author response:

Thanks for the suggestion. We have expanded the Related Work section by adding recent studies from 2025 and 2026 on DNN inference deployment, energy-aware edge inference, DAG-aware scheduling, reliability-aware edge computing, and multi-objective optimization. The revised discussion clarifies the gap addressed by this paper: dynamic DAG-aware deployment under joint service reliability, inference accuracy, inference delay, and total energy consumption.

Author action:

We revised Section 2 and updated the reference list. The revised Related Work section now explicitly compares existing studies with the proposed DAREED design from the perspectives of DNN topology, dynamic reliability, link reliability, energy consumption, and preference-based multi-objective deployment.

Reviewer#2, Concern#2: The authors assume an oversimplistic constant service reliability of computing nodes during execution, which is impractical in real systems.

Author response:

Thanks for the comment. We agree and have revised the model. Node service reliability is no longer constant. It is dynamically updated according to node load and heat state. This better reflects mobile edge environments where overloaded or overheated nodes may become less stable during long-running inference service.

Author action:

We added dynamic service reliability and dynamic inference accuracy equations in Section 4 and added dynamic degradation experiments in Section 6.

Reviewer#2, Concern#3: The authors only consider reliability of computing nodes while ignoring the reliability of communication links.

Author response:

Thanks for this important suggestion. The revised model includes communication link reliability. Link reliability changes with link load and heat state, and the service reliability of a DNN request considers both the computing nodes and communication links traversed by input uploading, intermediate dependency transmission, and output returning.

Author action:

We revised the reliability model in Section 4 to include dynamic link reliability and used-link sets in service reliability calculation.

Reviewer#2, Concern#4: The simulation environment is quite small and limited to only 10 edge nodes and 1 central cloud. It is suggested to test the model with larger network topologies.

Author response:

Thanks for the suggestion. We have added a large-scale network evaluation. The revised experimental design includes small, medium, and large network scenarios. .

Author action:

We added the “Large-scale Network Evaluation” subsection in Section 6.

Reviewer#2, Concern#5: The time complexity is quite high and will degrade at larger scale.

Author response:

Thanks for the comment. We have revised the algorithm design to include complexity analysis and scalability-oriented mechanisms. DAREED limits exhaustive search by using DAG-aware initialization, resource-aware mutation, constraint repair, and elite local search. These mechanisms reduce invalid exploration and guide the population toward feasible regions. We also added runtime reporting in the scalability experiments.

Author action:

We revised Section 5 to include complexity analysis and revised Section 6 to report running time under different request loads and network scales.

Reviewer#2, Concern#6: The proposed framework assumes a static network environment, while practical network conditions are dynamic.

Author response:

Thanks for the suggestion. The revised framework is explicitly dynamic. DNN requests arrive online. Accepted requests become running blocks and occupy node and link resources for multiple time slots. Node load, link load, heat state, effective bandwidth, node reliability, link reliability, and inference accuracy are updated as the system evolves.

Author action:

We revised Section 4 to include dynamic state transition `model(*,*,*)` and revised Section 5 to explain online execution of DAREED.

Reviewer#2, Concern#7: The paper only considers reliability and delay as optimization objectives, but does not consider energy consumption or cost.

Author response:

Thanks for the suggestion. We have incorporated **total energy consumption** as an optimization objective. The revised energy model includes computation energy and transmission energy. DAREED uses energy utility in candidate scoring, mutation guidance, local search, and final deployment selection. We did not include monetary cost as a primary objective in this revision because cloud-edge-end pricing models vary significantly across platforms, while energy consumption is directly measurable and closely related to resource efficiency.

Author action:

We revised Section 4 to add the total energy consumption model and objective. We revised Section 5 and Section 6 to include energy-aware algorithm mechanisms and energy-related performance metrics.

Response to Reviewer #3

Reviewer#3, Concern#1: RDODA is essentially a direct application of classic NSGA-II using standard genetic operations. It lacks customized algorithmic designs tailored to DNN graph dependencies or edge networks.

Author response:

Thanks for the critical comment. We have redesigned the algorithm and renamed it DAREED. The revised algorithm is no longer a direct use of standard NSGA-II. DAREED incorporates DNN topology and dynamic resource states into the whole evolutionary process. [Customized algorithmic designs tailored to DNN graph dependencies or edge networks](#). Specifically, it builds DAG context information, initializes the population with topology-aware and resource-aware modes, performs DAG block crossover, applies resource-aware skew mutation, repairs infeasible individuals, uses constraint-dominance sorting, refines elite individuals through local search, and selects the final deployment according to request preferences and energy cost.

Author action:

We rewrote Section 5 as “Algorithm Design” for DAREED and added detailed descriptions of each customized mechanism.

Reviewer#3, Concern#2: The strategy of assigning negative fitness values to individuals violating the delay constraint is overly simplistic and may destroy population diversity.

Author response:

Thanks for the suggestion. We replaced the previous penalty treatment with a constraint-aware mechanism. DAREED evaluates constraint violation separately from objective values. Feasible individuals dominate infeasible ones, and infeasible individuals with smaller violations are retained when useful for guiding search. This approach maintains population diversity while reducing the probability of selecting severely infeasible solutions.

Author action:

We revised Section 5 to include constraint-aware repair and constraint-dominance sorting.

Reviewer#3, Concern#3: The final selection algorithm uses a simple linear weighted sum, which degrades the multi-objective optimization back into a single-objective problem. If weights are known, why compute the Pareto front?

Author response:

Thanks for raising this important point. In the revised DAREED framework, Pareto search and final selection play different roles. The evolutionary stage generates a diverse non-dominated candidate set without fixing objective weights during candidate generation. The preference-based selection stage is then applied only after the candidate set has been obtained, so different DNN requests can select different deployment schemes from the same type of Pareto representation according to their service preferences. This preserves multi-objective exploration while allowing concrete online deployment decisions.

Author action:

We revised Section 5 to clarify the two-stage framework of evolutionary candidate search and preference-based deployment selection. We also added application preference experiments in Section 6.

Reviewer#3, Concern#4: Assuming that service reliability remains constant during task execution is highly unrealistic for mobile edge environments.

Author response:

Thanks for the comment. We agree and have revised the model. The revised model represents service reliability as a dynamic variable affected by node load and heat. Since accepted DNN requests occupy resources for several time slots, subsequent deployment decisions are made under updated reliability and resource states.

Author action:

We revised Section 4 to include dynamic node reliability and dynamic state transition, and we added dynamic degradation experiments in Section 6.

Reviewer#3, Concern#5: The reliability calculation multiplies assigned node reliabilities and assumes independent subtask failures. If multiple layers are deployed on the same node, failure of that node causes all subtasks to fail simultaneously.

Author response:

Thanks for pointing out this logical issue. We have revised the reliability formulation. The service reliability of a DNN request is now calculated over unique used computing nodes and traversed communication links. Thus, a node used by multiple DNN layers contributes only once to service reliability, which better captures correlated failures among layers deployed on the same node.

Author action:

We revised the service reliability model in Section 4 and updated the objective formulation.

Reviewer#3, Concern#6: Comparing the proposed method only against simple heuristics is unfair. The algorithm should be evaluated against recent state-of-the-art multi-objective optimization algorithms or learning-based deployment strategies.

Author response:

Thanks for the suggestion. We have expanded the baseline set. In the revised experiments, DAREED is compared with a generic NSGA-II based evolutionary deployment algorithm, simulated annealing, random feasible deployment, maximum-resource deployment, local-first deployment, and RTBL, which is a reliability-aware task scheduling method based on Lyapunov optimization and bandit learning. This allows us to evaluate both Pareto-front quality and online deployment performance against stronger baselines.

Author action:

We revised Section 6 to add comparison algorithms and separate Pareto-front quality evaluation from final deployment performance evaluation.

Reviewer#3, Concern#7: The experimental setup with ten edge nodes and one cloud node is too small to represent realistic large-scale edge computing scenarios. The high theoretical complexity raises scalability concerns.

Author response:

Thanks for the comment. We have added scalability experiments under small, medium, and large topologies. The large topology includes 1 cloud node, 60 edge nodes, and 300 end devices. We also report running time to evaluate whether DAREED remains practical as network scale and request load increase.

Author action:

We added large-scale network evaluation and runtime reporting in Section 6. We also revised Section 5 to discuss algorithm complexity.

Reviewer#3, Concern#8: The simulation relies on static parameter ranges and does not incorporate dynamic factors such as bandwidth fluctuations, node mobility, or background traffic variations.

Author response:

Thanks for the suggestion. We have revised the simulation model to include dynamic network and node states. Effective bandwidth changes with link load, while link reliability changes with link load and heat. Node reliability and inference accuracy also change with node load and heat. Although explicit mobility modeling is left for

future extension, the revised model captures the dynamic resource and reliability fluctuations caused by online workloads and background contention.

Author action:

We revised Section 4 to include dynamic bandwidth, dynamic link reliability, node/link load, heat evolution, and state transition. We added dynamic degradation and cloud-side stress experiments in Section 6.